HARSHIT VERMA

+91 6306050027 | varma789ajay@gmail.com | LinkedIn | GitHub

About Me

Motivated Computer Science undergraduate and GenAI Intern Trainee at WNS with hands-on experience in building AI-powered automation systems and intelligent document-processing solutions. Skilled in Python, LangChain, LangGraph, FastAPI, SQL, and React, with strong interest in Agentic AI, LLM applications, and workflow automation. Passionate about developing scalable AI solutions that improve efficiency, reduce manual effort, and solve real-world business problems.

Experience

Intern Trainee — GenAI & Automation *Mar 2026 – Present*

WNS Global Services (Holdings) Ltd.

- Architected a multi-agent AI pipeline (LangChain + LangGraph) to extract and validate structured fields from banking/insurance PDFs with per-field confidence scoring (≥85%) and auto-escalation on exceptions.
- Built production FastAPI backend: PDF upload, PyMuPDF/pdfplumber parsing, rules-engine validation, and audit-trail logging for governance compliance.
- Integrated a live AI chatbot (OpenRouter LLM) into a React + Vite frontend for interactive invoice querying with explainable AI outputs.

Projects

Invoice Validation Chatbot — Agentic AI GitHub *Mar 2026 – Present*

Python · FastAPI · LangChain · LangGraph · React · OpenRouter · PyMuPDF

- Architected a 3-agent LangGraph pipeline: Extraction Agent parses invoice fields (Invoice No, Dates, Addresses, Borrower details) via LLM with per-field confidence scores; Validation Agent enforces mandatory field checks, date format rules, and cross-field consistency with a confidence threshold of ≥85%.
- Developed an Escalation Agent that automatically flags low-confidence fields, missing values, and cross-field mismatches, triggers human-in-loop review for exceptions, and maintains a full audit trail for compliance and traceability.
- Integrated a live AI chatbot interface enabling natural-language querying of extracted invoice data, delivering explainable AI outputs and supporting straight-through processing (STP) across banking and insurance use cases.
- Built a React + Vite frontend with PDF preview and a real-time extraction results panel featuring color-coded confidence indicators (red/yellow/green) for each field to aid analyst decision-making.
- Eliminated manual data keying from documents, reducing processing errors and enforcing automated rule validation; system architected to scale across invoice, banking PDF, and insurance document workflows.

Attendance Management System — Face Recognition GitHub *Dec 2024 – Jan 2025*

Python · OpenCV · dlib · SQL · Tkinter

- Achieved 98% facial recognition accuracy using OpenCV and dlib's 68-point facial landmark model; automated real-time attendance marking for 100+ students per session, reducing manual effort by 80%.
- Designed an SQL-backed attendance logging system with automatic timestamping, supporting CSV and Excel export in under 1 minute for seamless integration with institutional record-keeping workflows.
- Built a Tkinter GUI featuring a live camera feed, real-time face detection overlays, and a manual override panel for faculty to correct or add attendance entries without disrupting the automated pipeline.
- Implemented a face encoding registration module allowing faculty to enroll new students by capturing multiple face samples, improving recognition robustness under varying lighting and angle conditions.

Skills

AI / GenAI: LangChain, LangGraph, OpenRouter API, LLM Prompt Engineering, RAG, Agentic AI Pipelines

Languages & Frameworks: Python, Java, JavaScript, SQL · FastAPI, React (Vite), Spring Boot, OpenCV, pandas, numpy

Tools & Databases: GitHub, Postman, VS Code · SQLite, MySQL, Firebase · PyMuPDF, pdfplumber, pytesseract

Education

B.Tech — Computer Science & Engineering *2022 – 2026*

A.P.J. Abdul Kalam Technical University, Lucknow

Certifications & Achievements

- Programming in Python — Coursera (Oct 2023) | Adobe India Hackathon — Certificate of Participation (Aug 2024)
- Organized Software Testing Workshop for 50+ participants | National-level Data Science Hackathon participant